

Natural Language Processing With The Python NLTK

>>The technology behind all the soulful conversations you've had with your phone> by **Kshitij Sobti**

Computers have constantly been evolving to be easier to use and understand. They are now powerful enough to try and understand humans, rather than needing humans to understand them.

Apple's Siri, Microsoft's Cortana, Amazon's Alexa and Google Now are a new kind of interface that relies on asking your device to perform actions, or retrieve data using natural speech.

Of course the first step these digital assistants do is to convert the voice data to text using speech recognition. But what then?

When you ask Google Now, "What is the cost of the cheapest flight to Sydney leaving August 11th", after converting that to text it still needs to actually recognise that you are querying flights from your current location to Sydney; that you want to travel on August 11th and that you want the cheapest flight. Traditionally this would involve filling in complex forms.

Furthermore you could easily ask: "Which is the cheapest flight to Sydney on August 11th" or just "cheapest Sydney flight august 11th" and you should get the same result. Clearly Google is able to parse the key bits of data in your query and actually act on it.

Natural Language Processing

Natural Language Processing (NLP) is the field of Computer

Science that deals with developing systems that can work with natural language.

While most programming languages and markup is text data that computers need to process, NLP is fundamentally different since most natural languages don't have a standardised unambiguous syntax. Natural languages like English can often be very ambiguous.

Consider the sentence, "We saw her duck." Does it reference the act of ducking. Does mean to say we saw a duck that belonged to this person. Or even that people used a saw on the duck to cut it. We sure hope it wasn't this last one.

Without additional context it's hard to know what this means. Given our extensive experience with the world, we can say that the first interpretation is most likely and the last one is least likely. After all few people have ducks around and even fewer (one can hope) would need to cut a duck using a saw. On the other hand people duck all the time. And with that knowledge we can reason that it was the first case that happened, unless told otherwise.

Computers can't intuit the most probable meaning of this sentence like humans can. Or can they? Using machine learning it is possible to train computers to better make predictions about the meaning of text. As you can imagine they need to digest a lot of

*Coding Matters



*Hotel search chatbot

>> DMI works with Zumata to create hotel search chatbot with integrated NLP.

<http://dgit.in/DMIZumata>



*Microsoft's NLP app

>> Microsoft launches CAP which is an IFTTT style natural language processing web app

<http://dgit.in/MSCAPnlp>



*Niki gets NLP upgrade

>> Niki, the chatbot that lets you order anything, gets advanced language processing abilities.

<http://dgit.in/NikiNLP>